SAEON Product Backlog

The ESP32 WROOM-32 reads the TR-525M rain gauge. DHT11 is used for testing; SHT31 is the temperature and humidity sensor selected for the final product. Firebase is the database platform.

PB01 Project Initiation & System Architecture

Week 1 | 28 July 2026 to 31 July 2026 | US01, US02, US03, US04, US13

Product Backlog	In Progress	Completed
1.01 Define Project Scope & Board Setup		Completed
1.02 Component Verification & Sourcing		Completed
1.03 Development Environment Configuration		Completed
1.04 Architecture Diagram & Data Flow		Completed

PB02 Core Firmware & Sensor Interfacing

Week 2 | 4 August 2026 to 7 August 2026 | US01, US02, US11

Product Backlog	In Progress	Completed
2.01 ESP32 Base Setup & Serial Output		Completed
2.02 DHT11 Testing and SHT31 Final Sensor Requirements		Completed
2.03 TR-525M Rain Gauge Logic		Completed
2.04 Firmware Formatting & Low Power Mode		Completed

PB03 HiveMQ Cloud Broker Setup

Week 3 | 11 August 2026 to 14 August 2026 | US03, US04, US09

Product Backlog	In Progress	Completed
3.01 Broker Provisioning		Completed
3.02 Topic Structuring & Access Control		Completed
3.03 Network Connection & Publish Testing		Completed
3.04 Firebase Storage Integration		Completed

PB04 Cellular Communication (SIM7000G)

Week 4 | 18 August 2026 to 21 August 2026 | US03, US04

Product Backlog	In Progress	Completed
4.01 SIM Module Wiring & Power		Completed
4.02 Network Bearer Configuration		Completed
4.03 Cellular MQTT to Firebase Integration		Completed
4.04 Stability & Error Handling		Completed

PB05 SAEON Dashboard Base Application

Week 5 | 25 August 2026 to 28 August 2026 | US05, US06, US09

Product Backlog	In Progress	Completed
5.01 App Skeleton & Routing		Completed
5.02 Real-Time Visualization		Completed
5.03 Historical Data & Filtering		Completed
5.04 Cloud Command Prep		Completed

PB06 Power Supply & Solar Integration

Week 6 | 1 September 2026 to 4 September 2026 | US11

Product Backlog	In Progress	Completed
6.01 Battery & Charge Controller Setup		Completed
6.02 Solar Panel Integration		Completed
6.03 Mains Power Alternative Testing		Completed
6.04 System Power Flow Audit		Completed

PB07 Advanced UI & Health Diagnostics

Week 7 | 8 September 2026 to 11 September 2026 | US07, US08, US10

Product Backlog	In Progress	Completed
7.01 Develop Station Health Diagnostics UI Component		Completed
7.02 Implement CSV Export Functionality		Completed
7.03 Mobile & Tablet Responsiveness		Completed
7.04 Full End-to-End Data Pipeline Ping Test		Completed

PB08 Hardware Finalization & PCB Soldering

Week 8 | 15 September 2026 to 18 September 2026 | US12

Product Backlog	In Progress	Completed
8.01 Circuit Transfer & Layout		Completed
8.02 PCB Soldering		Completed
8.03 Enclosure Preparation		Completed
8.04 Component Mounting & Sealing		Completed

PB09 Pre-Deployment Auditing & Calibration

Week 9 | 22 September 2026 to 25 September 2026 | US11, US12

Product Backlog	In Progress	Completed
9.01 Power Drain Audit		Completed
9.02 Solar Charging Verification		Completed
9.03 Full System Deployment	In Progress	
9.04 Baseline Calibration	In Progress	

PB10 48-Hour Field Stress Test

Week 10 | 29 September 2026 to 2 October 2026 | US01, US02, US03, US11, US12

Product Backlog	In Progress	Completed
10.01 Stress Test Initiation	In Progress	
10.02 Active Monitoring	In Progress	
10.03 Manual Field Measurements	In Progress	
10.04 Data Auditing & Conclusion	In Progress	

PB11 Presentation Formatting & Content

Week 11 | 6 October 2026 to 9 October 2026 | US13

Product Backlog	In Progress	Completed
11.01 Presentation Outlining		Completed
11.02 Technical Diagrams & Architecture Slides		Completed
11.03 Dashboard & Software Slides		Completed
11.04 Technical Documentation		Completed

PB12 Rehearsal & Submission Prep

Week 12 | 13 October 2026 to 16 October 2026 | US13

Product Backlog	In Progress	Completed
12.01 Slide Deck Finalization		Completed
12.02 Group Rehearsal 1		Completed
12.03 Timed Rehearsal & Feedback		Completed
12.04 Final Compilation		Completed

PB13 Project Buffer & Final Review

Week 13 | 20 October 2026 to 23 October 2026 | US12, US13

Product Backlog	In Progress	Completed
13.01 Buffer & Prototype Documentation Sign-off		Completed